

Hydrogen-enriched combustion via exhaust-driven ammonia reforming: A pathway to high-efficiency gas turbine operation

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ABSTRACT

In the quest for sustainable energy solutions, CO₂-free power generation has become increasingly important. Ammonia has emerged as a promising alternative fuel for gas turbines, offering a pathway toward decarbonization and reduced emissions. This paper explores on-board hydrogen production from ammonia via thermochemical decomposition using exhaust heat from a gas turbine. To demonstrate the synergy between gas turbine exhaust heat recuperation and on-board hydrogen production, the efficiency of a chemically recuperated combined cycle gas turbine (CRCCGT) was compared to that of a conventional combined cycle gas turbine (CCGT) through thermodynamic analysis. The analysis was conducted over a wide range of operational parameters. Results show that the thermal efficiency of the CRCCGT and its potential for hydrogen production depend on turbine inlet temperature and pressure. The CRCCGT achieves higher thermal efficiency than the CCGT at the optimal pressure ratio, with a peak value of 0.476. The integration of ammonia reforming via thermochemical decomposition with exhaust heat recuperation allows for the replacement of ammonia with hydrogen-rich gas (containing up to 75% hydrogen), significantly improving thermal efficiency. Additionally, the flame speeds of ammonia and hydrogen-rich gas were compared for various compositions, demonstrating the beneficial effect of ammonia decomposition prior to combustion on flame speed.

1. Introduction

During the next 30 years, the amount of yearly produced ammonia is predicted to increase by around 30% [1]. According to the sustainable development scenario, the share of NH₃ produced with near-zero emissions will increase up to 66% [2]. These prospects in NH₃ production development make NH₃ a very promising fuel in terms of transition to clean energy. Ammonia produced from renewable energy sources has recently received international interest for power generation [3–6]. Ammonia is an example of carbon-free fuel, and its combustion does not produce CO₂. Some successful applications use the NH₃ as fuel for power generation. The 50 kW class gas turbine with NH₃-air combustion was analyzed by Kurate et al. [7]. The 2 MW gas turbine was tested by the IHI corporation [8], the 40 MW gas turbine was developed by Mitsubishi power [9], and more gas turbines could be found in other companies [10–12].

When NH₃ is used directly as a gas turbine fuel, the turbine suffers from low combustion efficiency due to the low flame speed and long quenching distance of the fuel [13]. Also, NH₃ has a lower heat of combustion per 1 kg than widely used methane by more than two times. The low laminar burning velocity of NH₃ leads to longer flames

with slow diffusion [14]. This, in turn, leads to the necessity of using a combustion chamber with a larger volume. Slow diffusion rate results in incomplete combustion of fuel and NH₃ emissions [15]. Ammonia fuel also has auto-ignition temperature and minimal ignition energy higher than that of natural gas [16]. These properties are undesirable as the ignition process becomes harder, and ammonia flames are unstable with a tendency to blow off.

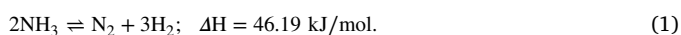
To mitigate the deficiencies of ammonia fuel in gas turbines, it is mixed with other fuels [17,18]. The addition of methane or hydrogen to ammonia improves the laminar burning velocity, decreases ignition temperature and increases the heat release rate [19]. The addition of H₂ to ammonia is more perspective as the lean/rich combustion limits become wider and blow-out limits are lower compared to the methane–ammonia mixture [20]. Also, the higher initial temperature of the fuel further increases the range of both lean and rich combustion [21]. The combustion of H₂ does not produce CO₂ and is considered more environmentally friendly than the combustion of CH₄ [22].

The use of H₂ addition to ammonia is beneficial from a combustion perspective [23]. However, the production and transportation of H₂ is very complex and has a high cost [24]. The use of dual fuel supply

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system (e.g. ammonia and hydrogen) in ammonia-fed power generation system requires additional resources and increases capital expenditure for power generation system. Hydrogen could be sustainably produced using water and air, but significant additional costs and complicated transportation systems are required for H_2 production using this technology near the power plant [25]. Another way is the transportation of H_2 to the power plant, which is also very expensive. Additional measures during transportation are also required to prevent leakage and explosion incidents [26]. Moreover, the storage cost of hydrogen (14.95 \$/kg H_2) is significant compared to that of ammonia (0.54 \$/kg H_2) in the course of 182 days [27]. However, some authors recently paid attention to H_2 production on-site via the fuel reforming [28–30]. This way of H_2 production is called on-board hydrogen production.

The mixing of ammonia and hydrogen is a promising pathway since H_2 can be obtained from thermocatalytic cracking of NH_3 , eliminating the need to store two fuels [28,31,31]. Instead of a whole additional fuel supply system for H_2 , this technology requires only minor additions — fuel reformer before combustion. The ammonia decomposition is an endothermic reaction that occurs in the vicinity of the metal catalyst. The ammonia decomposition reaction is written as:



The lower heating value (LHV) of reaction products is 1.14 times greater than LHV of ammonia due to endothermic nature of reaction (1). These reaction products are H_2 -rich gas that can be directly fed to the combustor of gas turbine.

The dissociation of NH_3 has a strong dependence on the temperature, pressure, and catalytic reactivity of the reformer. However, the minimum temperature requirement for an almost complete decomposition is approximately 430 °C under atmospheric pressure [32]. The exhaust gases of NH_3 -fueled gas turbine can cover this temperature requirement even at relatively low equivalence ratio combustion conditions. Previous studies have reported that ammonia-fueled gas turbines can produce exhaust gas temperatures of over 500 °C [33,34], providing sufficient thermal energy for H_2 generation.

The exhaust gas temperature after ammonia decomposition is still high and may exceed 400 °C [28]. Thus, the heat of exhaust gases could be further used to produce steam. The generated steam is used in the steam turbine and increases the net power output and efficiency of the gas turbine scheme. A thermodynamic analysis is widely used to properly compare the characteristics of various combined cycle gas turbine schemes. Ezzat et al. [35] used thermodynamic analysis to study exhaust heat recuperation in a gas turbine fueled by ammonia. The exhaust heat was used to produce electricity via the Rankine cycle, as well as hydrogen, hot water, and refrigeration. Selvam et al. [36] thermodynamically compared the performance of four bottoming steam cycles in a solid oxide fuel cell system. Verkhivker and Kravchenko [37] performed the thermodynamic analysis of a combined cycle power plant with chemical recuperation via steam methane reforming reactions.

This paper investigates the synergistic integration of ammonia reforming for on-board hydrogen production and gas turbine exhaust heat recuperation. The proposed system leverages two key interactions: first, the endothermic decomposition of ammonia, driven by waste heat from the exhaust stream, generates a hydrogen-rich reformat gas with superior combustion characteristics — including higher flame speed and greater stability — compared to pure ammonia. Second, the utilization of exhaust energy (at temperatures exceeding 400 °C) to supply the reaction enthalpy significantly improves the cycle's energy efficiency by upgrading low-grade thermal energy into chemical energy stored in the fuel. The core objective of this study is to quantify the benefits of this coupled process through a comparative thermodynamic analysis. The thermal efficiency of a gas turbine system incorporating on-board hydrogen production via exhaust-driven reforming is evaluated and compared to a baseline case fueled directly with ammonia. Additionally, the laminar flame speed of the resulting hydrogen–nitrogen mixture is computed to assess the potential improvements in combustor performance.

2. Methodology

Currently, the utilization of ammonia as a fuel for gas turbines is primarily centered on its combination with other gases such as hydrogen, methane, or synthesis gas. This approach is adopted due to the inherent limitations of pure ammonia, including its low combustion efficiency and stability. Additionally, the low flame speed of ammonia requires a larger combustion volume that demands modifications to the combustion chambers of gas turbines. On the positive side, ammonia has found great interest as a hydrogen carrier because it can be easily converted to hydrogen-rich gas via its decomposition to H_2 and N_2 using external heat sources. For gas turbine application, exhaust heat can be considered as an external heat source using thermochemical recuperation principle [38–40]. Hydrogen-rich gas after ammonia decomposition has better combustion performances than pure ammonia. Therefore, it is expected that decomposing ammonia before combustion can significantly improve combustion performance. At the same time, the use of exhaust heat for thermochemical ammonia decomposition will increase the thermal efficiency of gas turbines [28].

2.1. Concept of thermochemical recuperation

Thermochemical exhaust heat recuperation concept is based on the use of exhaust heat for endothermic thermochemical primary fuel transformation before its combustion. In this case, the heat of exhaust gases is converted into chemical energy of the products of such thermochemical transformation. Thermochemical recuperation is usually used in industrial applications with high temperatures of exhaust gas [41]. Recently, a set of papers was published by many research groups on the use of thermochemical recuperation in industrial furnaces [41,42], internal combustion engines [43,44], etc. Exhaust gas after gas turbines has a reasonably high temperature (more than 500 °C in some cases) that allows to use the exhaust gas for thermochemical ammonia decomposition. Gas turbines with thermochemical recuperation (TCR) are usually referred to as chemically recuperated gas turbines (CRGT). The ammonia decomposition reaction (Eq. (1)) is endothermic. The reaction product of ammonia decomposition is H_2 -rich gas with combustion heat higher than that of initial ammonia. H_2 -rich gas obtained via decomposition of 1 mole of NH_3 has a higher lower heating value than unreformed ammonia by the amount of consumed heat ΔH (Eq. (1)). Also, H_2 production is a significant advantage as pure ammonia combustion suffers from low combustion characteristics such as low flame speed, high flammability limits and others. The addition of H_2 to ammonia fuel mitigates these disadvantages, thus improving fuel combustion.

The ammonia-fired CRGT is schematically shown in Fig. 1. The gas turbine has traditional parts and additional equipment for thermochemical recuperation — ammonia reformer and evaporator. The hot exhaust gas after the combustion chamber is used to decompose the gaseous ammonia in the reformer. After the reformer exhaust gas heat is utilized to transform the liquid ammonia into the gaseous state. The temperature for a close to complete ammonia decomposition is above 300 °C, which is greater than the temperature of gaseous ammonia before the reformer. Thus, in the reformer, ammonia heating occurs prior to decomposition reactions. The H_2 -rich gas after the reformer has a high temperature and is fed to the combustion chamber.

In traditional ammonia-fed gas turbine, the chemical energy of ammonia is converted into heat in a single stage via its combustion. At the same time, in ammonia-fed gas turbine the chemical of ammonia is converted into heat via two stages. The first stage is ammonia reforming into hydrogen-rich gas that has a lower heating value of 1.14 times greater than that for pure ammonia. The second stage is combustion of hydrogen-rich gas in a combustion chamber. It should be noted that the exhaust heat also uses to heat up reaction mixture. Therefore, a part of exhaust heat is used to compensate the endothermic effect of ammonia decomposition reaction and part of exhaust heat is used to change the

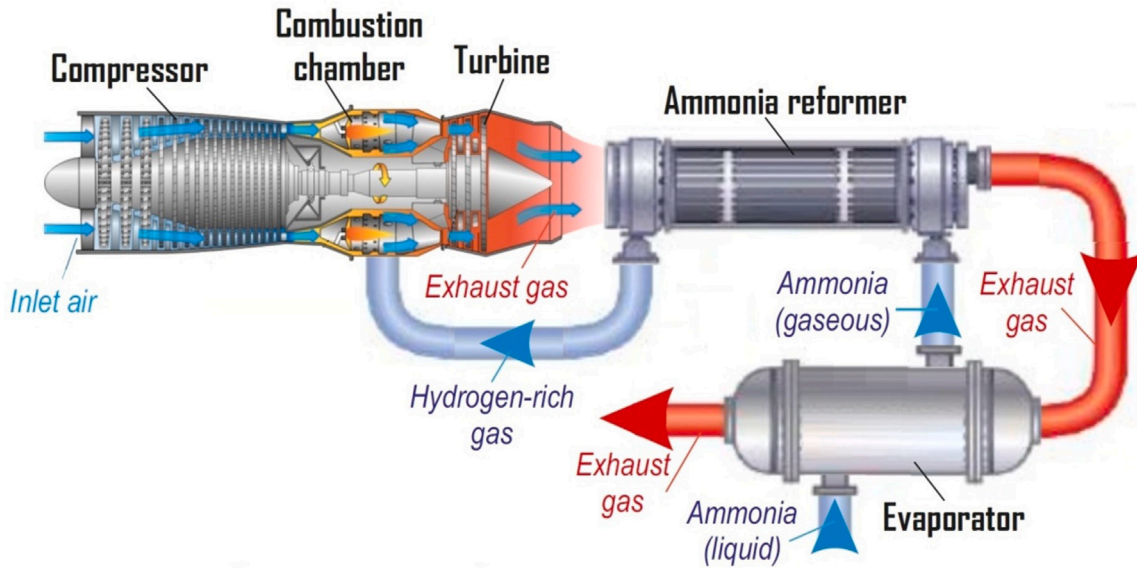


Fig. 1. The schematic diagram of an ammonia-fed chemically recuperated gas turbine.

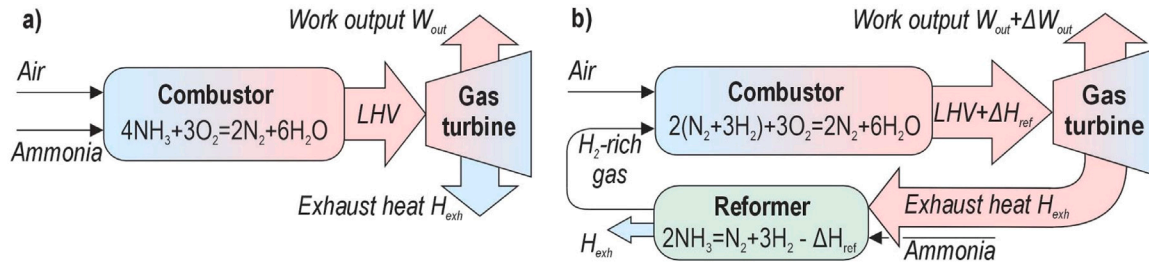


Fig. 2. Chemical energy transformation into heat in traditional ammonia-fed gas turbine (a) and in ammonia-fed gas turbine with thermochemical recuperation (b).

enthalpy of reaction mixture. The detailed analysis of energy balance is provided below. Thus, schematically the transformation of chemical energy of ammonia into heat for traditional ammonia-fed gas turbine and for ammonia-fed gas turbine with thermochemical recuperation via ammonia decomposition is presented in Fig. 2.

The exhaust gas after the reformer still (Fig. 2) has a high temperature (sometimes exceeding 300 °C), which could be used further. One of the effective ways to utilize the heat from the exhaust gases of a gas turbine is to install a steam turbine for bottoming cycle. Fig. 4 schematically shows the combined cycle gas turbine (CCGT) with the thermochemical exhaust heat recuperation system. The CCGT that includes TCR is a chemically recuperated combined cycle gas turbine (CRCCGT). In CRCCGT, part of the exhaust gas heat is initially used for NH_3 decomposition, then the cooled exhaust gas is used to produce steam for a steam turbine. In the present investigation, the prospects of CRCCGT were assessed and compared to those of traditional combined cycle gas turbine (CCGT) (see Fig. 3).

2.2. Thermodynamic analysis

The thermodynamic analysis was performed in Aspen Hysys software, a powerful tool for designing and optimizing energy and chemical processes. Three gas turbine schemes were analyzed and compared: gas turbine cycle without recuperation, combined cycle gas turbine, and chemically recuperated combined cycle gas turbine. For each scheme a parametric analysis was performed to determine the dependence of thermal efficiency on operating parameters such as gas turbine inlet temperature and operating pressure. The choice of optimal

ammonia-fired gas turbine scheme based on operational parameters was discussed.

In this paper, three schemes were analyzed:

- scheme a: Simple Ammonia-fed Gas Turbine (GT) without recuperation (baseline);
- scheme b: Chemically Recuperated Gas Turbine (CRGT);
- scheme c: Simple Combined Cycle Gas Turbine (CCGT);
- scheme d: Chemically Recuperated Combined Cycle Gas Turbine (CRCCGT).

It should be noted that all three schemes are fueled with ammonia.

The computational scheme for investigated gas turbines is shown in Fig. 4. The gas turbine schemes include an air compressor, combustion chamber, gas turbine and steam turbine. The main difference of the CRCCGT scheme in Fig. 4c is the use of exhaust gas heat for ammonia decomposition prior to the steam turbine cycle. The advantages of thermochemical recuperation (TCR) via ammonia decomposition were discussed earlier in this section. The addition of a steam turbine after the TCR system allows the utilization of exhaust heat after the NH_3 decomposition reformer. Thus, the net work of the gas turbine increases. However, the positive effect of steam turbine addition insignificantly decreases due to the water pump, which requires work. The schemes were compared for the same gas turbine inlet temperature. The adjust function was used to control the gas turbine inlet temperature due to a change in air mass flow rate.

The thermodynamic analysis of simple ammonia-fed gas turbine with thermochemical recuperation and ammonia-fed combined cycle

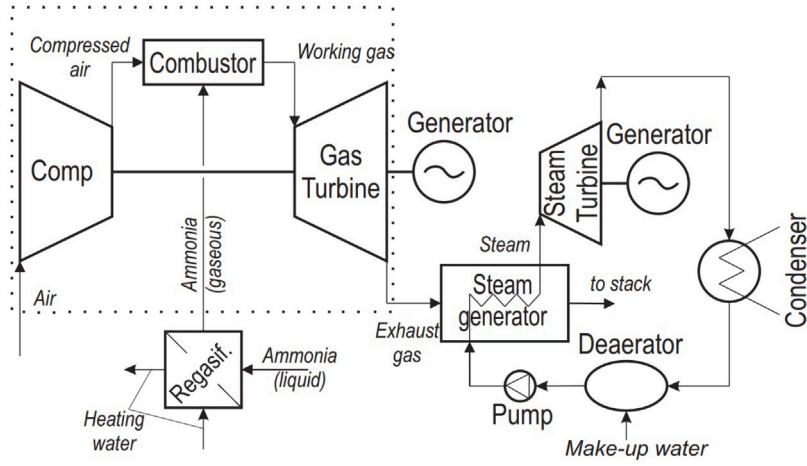


Fig. 3. Schematic diagram of combined cycle chemically recuperated gas turbine.

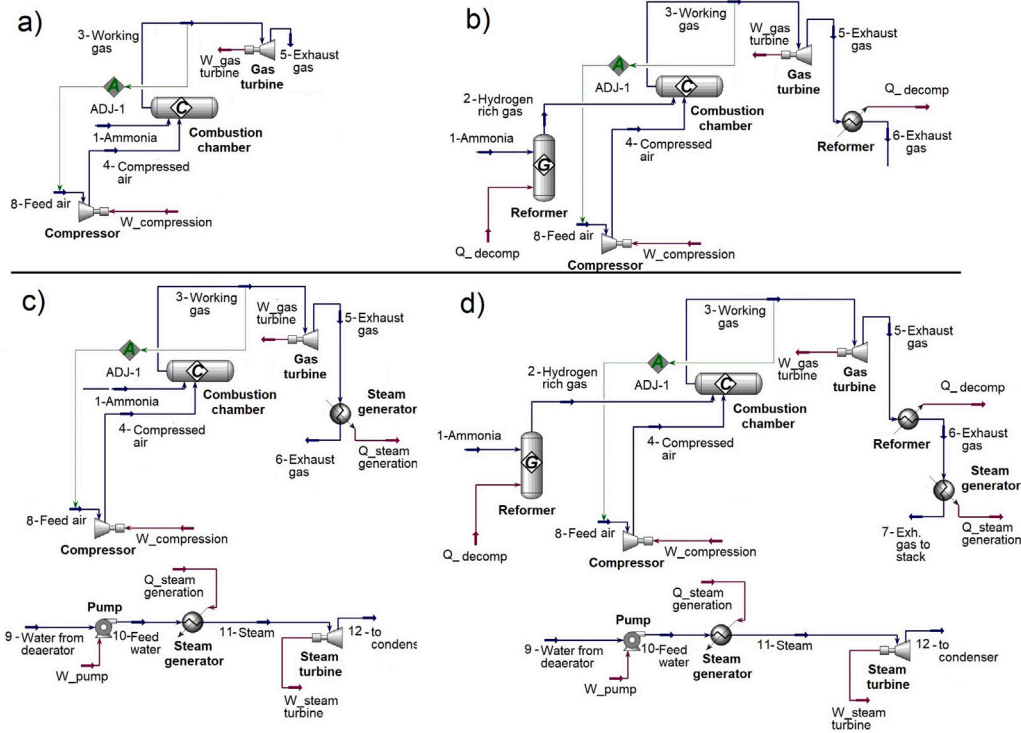


Fig. 4. Computational schemes for Simple Ammonia-fed Gas Turbine (GT) without recuperation (a); Chemically Recuperated Gas Turbine (CRGT) (b); Simple Combined Cycle Gas Turbine (CCGT) (c); Chemically Recuperated Combined Cycle Gas Turbine (CRCCGT) (d).

gas turbine where the exhaust heat is used only for steam turbine cycle was performed for comparative analysis (Fig. 4a).

The net work of gas turbine was found according to the expression:

$$W_{net} = W_{GT} + W_{ST} - W_{comp} - W_{pump} \quad (2)$$

where W_{GT} is the gas turbine power output; W_{ST} is steam turbine power output; W_{comp} is power consumed by air compressor; W_{pump} is power consumed by water pump. The W_{ST} and W_{pump} will equal to zero in gas turbine scheme without the steam turbine.

To compare the performance of different gas turbine schemes, the gas turbine thermal efficiency was chosen. The thermal efficiency is written as:

$$\eta_{GTU} = \frac{W_{net}}{m_{fuel} \cdot LHV_{fuel}} \quad (3)$$

where m_{fuel} is the mass of fuel and LHV_{fuel} is the lower heating value of the fuel. In a gas turbine without recuperation, the fuel is ammonia gas, while in a gas turbine with TCR, it is a H_2 -rich gas mixture.

Thermal efficiency was assessed for gas turbine inlet temperature in the range of 800–1400 °C and operating pressure in the range of 5–25 bar. Various gas turbine schemes could be adequately compared with equal gas turbine inlet temperatures. Gas turbine inlet temperature is heavily influenced by the type of fuel used in the combustion chamber. Two types of fuel are present in this investigation: the unreformed NH_3 and H_2 -rich fuel - a product of ammonia decomposition reaction. The combustion of NH_3 and H_2 -rich gas have various adiabatic flame temperatures. To sustain equal temperature at the inlet of the gas turbine, the air mass flow was changed.

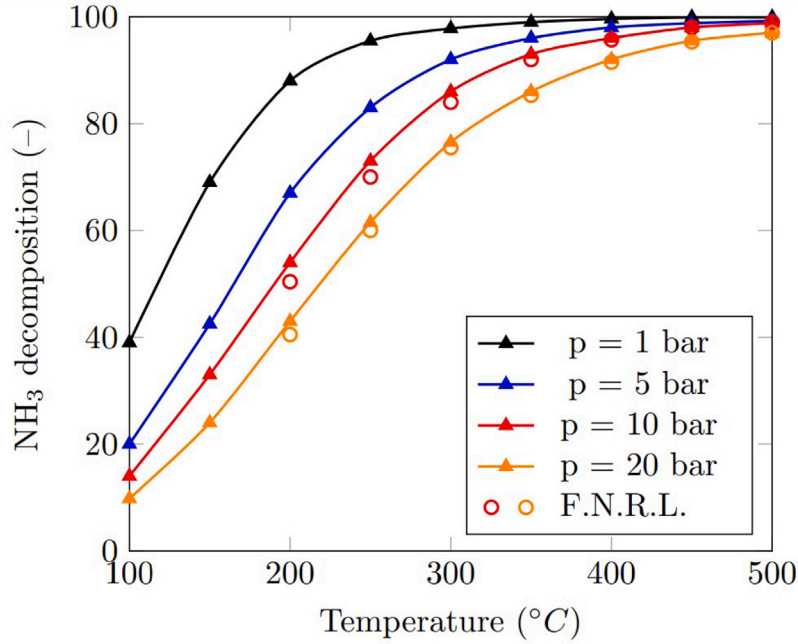


Fig. 5. Comparison of ammonia decomposition under equilibrium conditions obtained from numerical modeling (solid lines) and experimental data (hollow points) from Fixed Nitrogen Research Laboratory (F.N.R.L.) [45].

The fuel composition after NH_3 reforming varies significantly from temperature and pressure. A thermodynamic analysis was used to determine the composition of ammonia decomposition reaction products. Thus, the thermodynamic analysis was conducted for various gas turbine schemes and the reaction of ammonia decomposition. Thermodynamic analysis allowed to calculate the equilibrium composition of the reaction products for given operating conditions. The products of the ammonia decomposition reaction were determined by Gibbs's free energy minimization method. The main concept of the Gibbs free energy minimization method states that the system is in a thermodynamically favorable state when Gibbs free energy is minimum and its differential is zero:

$$dG^t = 0 \quad (4)$$

where G^t is Gibbs free energy.

The sum of the chemical potential of all ammonia decomposition reaction components yields the total Gibbs free energy of the reaction:

$$G^t = \sum \mu_i n_i \quad (5)$$

where n_i is stoichiometric coefficient of i th component and μ_i is a chemical potential of i th component.

The chemical potential of each component is calculated as:

$$\mu_i = \Delta G_{f_i}^0 + RT \ln\left(\frac{f_i}{f_i^0}\right) \quad (6)$$

where $G_{f_i}^0$ is the standard free energy of i th component formation; R is the universal gas constant; T is operating temperature; f_i is fugacity of a pure i th component and f_i^0 is fugacity of a pure i th component in standard state.

The equation of total Gibbs free energy of a chemical system could be obtained by combining Eqs. (4)–(6) in the following form:

$$G^t = \Delta G_{f_i}^0 + \sum n_i RT \ln\left(\frac{f_i}{f_i^0}\right) \quad (7)$$

The Peng–Robinson equation was used to calculate the volume of ammonia decomposition reaction products for various temperatures and pressures.

Thermodynamic analysis is a convenient tool for calculating the maximum thermal efficiency of a gas turbine and the equilibrium

composition of reforming products for specific operating conditions. However, the thermodynamic analysis presented in this paper had several limitations: it did not account for pressure drop in the system, a steady state process, the heat flux through the reactor wall was absent, and chemical reaction kinetics was at equilibrium.

3. Results

3.1. Verification

The thermodynamic analysis was used in this paper to obtain the results of the NH_3 decomposition reaction. The verification was performed to ensure the accuracy of numerical modeling results. The comparison of results obtained from thermodynamic analysis via Aspen Hysys with experimental results from Fixed Nitrogen Research Laboratory (F.N.R.L.) is shown in Fig. 5. The values in Fig. 5 are presented for equilibrium conditions inside the chemical reactor. The characteristic dependence is observed: high temperature and low operating pressure are favorable for NH_3 decomposition. This behavior is consistent with Le Chatelier's principle. The experimental and numerical results lie in close agreement with a maximum deviation of 5.7% for pressure 10 bar and temperature 200 °C. This comparison confirms the reliability of the obtained results.

3.2. Thermodynamic analysis

The results of thermodynamic analysis can be divided into two stages. The first part is dedicated to studying the chemical process inside the reformer. The NH_3 decomposition reaction inside the reformer depends heavily on operating conditions such as temperature and pressure. Therefore, it is important to study it separately to understand the limits of NH_3 decomposition for given operating conditions. The second part of the results compares the thermal efficiency of three computational schemes: gas turbine without recuperation; combined cycle gas turbine; chemically recuperated gas turbine with a steam turbine. The thermal efficiency of each computational scheme varies depending on operating conditions, and it is of great interest to determine which one performs better for a given set of parameters.

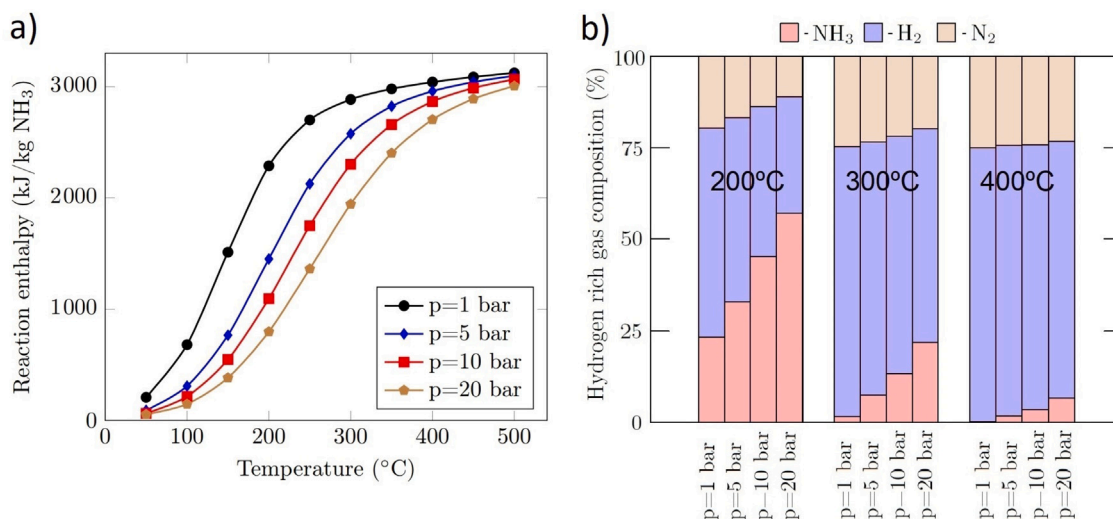


Fig. 6. (a) Reaction enthalpy and (b) composition of ammonia decomposition reaction products for various reformer temperatures and pressures.

The characteristics of NH₃ decomposition reaction depend heavily on reactor operating temperature and pressure. The enthalpy of NH₃ decomposition reaction (Eq. (1)) with respect to reactor temperature and pressure is shown in Fig. 6a. The enthalpy values are calculated for the reaction equilibrium state inside the reactor. Thus, the reaction enthalpy allows determining the heat consumption limit by the endothermic NH₃ decomposition reaction per kg of NH₃ under given operating conditions. As the temperature inside the reactor increases, the reaction enthalpy increases rapidly in the range between 100–300 °C and then increases slowly. Under higher operating pressure, the reaction enthalpy decreases. The most significant enthalpy difference of 64 % is observed between 1 bar and 20 bar for a reactor temperature of 200 °C. Therefore, for application with a reactor operating in the temperature range below 300 °C, operating pressure inside the reactor is one of the critical factors.

The NH₃ decomposition reaction product is H₂-rich gas. The higher the H₂ fraction in the H₂-rich gas, the higher the lower heating value of the fuel. The composition of H₂-rich gas for various operating pressures and temperatures inside the reformer is shown in Fig. 6b. Higher temperature and lower pressure have a positive influence on H₂ content. The temperature increment from 200 °C to 400 °C increases the H₂ mole fraction nearly twice for high operating pressure (10 bar and 20 bar). For a temperature of 400 °C, nearly all NH₃ is converted. Therefore, a further increase in reactor temperature will have little positive effect on the NH₃ decomposition process. The increase in pressure from 1 bar to 20 bars decreases the H₂ molar fraction by nearly 25 % for temperatures of 200 °C and 300 °C. However, the effect of operating pressure for 400 °C becomes less pronounced.

The results of thermodynamic analysis of various computational schemes are discussed in terms of thermal efficiency. The thermal efficiency of ammonia-fed gas turbine with on-board hydrogen production via thermochemical exhaust heat recuperation as a function of turbine inlet temperature and pressure is presented in Fig. 7b. Fig. 7b shows that for high temperature gas turbines (e.g. G-series, J-series) with turbine inlet temperature above 1400 °C, the thermal efficiency can reach up to 44%. Comparing the results from Fig. 7b for CRGT (scheme b) and the results from Fig. 7a for simple ammonia-fed gas turbine (scheme a), it can be concluded that the use of on-board hydrogen production from ammonia via thermochemical recuperation increases the thermal efficiency of gas turbine up to 9%. The main reason for this increase is the fact that exhaust heat is recuperated into gas turbine cycle decreasing heat losses with exhaust gases. Therefore, the first point of the synergistic effect of on-board hydrogen production via

ammonia decomposition and gas turbine exhaust heat recuperation is the higher thermal efficiency of gas turbine.

To better understand the synergistic effect of on-board hydrogen production via ammonia decomposition and gas turbine exhaust heat recuperation, the combined cycle gas turbines with and without thermochemical recuperation were thermodynamically analysed.

The thermal efficiency of gas turbines without recuperation and combined cycle gas turbines are presented in Fig. 7a and b, respectively. The thermal efficiency of a combined cycle gas turbine is 12%–20% higher than that of a gas turbine without recuperation. The optimal pressure ratio for a combined cycle gas turbine is shifted to the left (a point with a lesser pressure ratio). The optimal pressure ratio for a gas turbine without recuperation is 15 bar for 800 °C and 24 bar for 1000 °C. The highest thermal efficiency for a combined cycle gas turbine is observed for 12 bar for 800 °C and 21 bar for 1000 °C.

The comparison shown in Fig. 7 clearly demonstrates the increase in gas turbine thermal efficiency due to the implementation of exhaust heat recuperation. Steam turbines are widely used in gas turbine applications to benefit from exhaust heat recuperation. Therefore, the combined cycle gas turbine may be used as a benchmark case for further comparison.

The thermal efficiency of two computational schemes of gas turbines with various exhaust heat recuperation systems are compared in Fig. 8 - simple combined cycle gas turbine (scheme c) and chemically recuperated combined cycle gas turbine (scheme d). The computational scheme shown in Fig. 4d performs exhaust heat recuperation in two stages: high-temperature exhaust gas is used in an ammonia decomposition reactor, and then the inter-cooled exhaust gas is directed to steam generation for a steam turbine. The dependence character of thermal efficiency on temperature and pressure is very similar for both cases. The thermal efficiency for an optimal pressure ratio is higher for a gas turbine with chemical recuperation and a steam turbine for all inlet temperatures. For the scheme d in Fig. 8a the highest thermal efficiency is 0.372 for 800 °C and 0.411 for 1000 °C. For the scheme c in Fig. 8b, the highest thermal efficiency is 0.357 for 800 °C and 0.394 °C for 1000 °C. The main reason for the higher thermal efficiency of gas turbines with chemical recuperation and steam turbines is the use of chemically recuperated heat for work output in gas turbines, which results in higher efficiency. However, it should be noted that the process inside the chemical reactor is subject to lower reaction rates due to the reactor design, catalyst type, catalyst mass, reactant mass flow, heat losses and others. Nevertheless, with the help of existent techniques, the reactor operation close to equilibrium conditions is achievable [46,47].

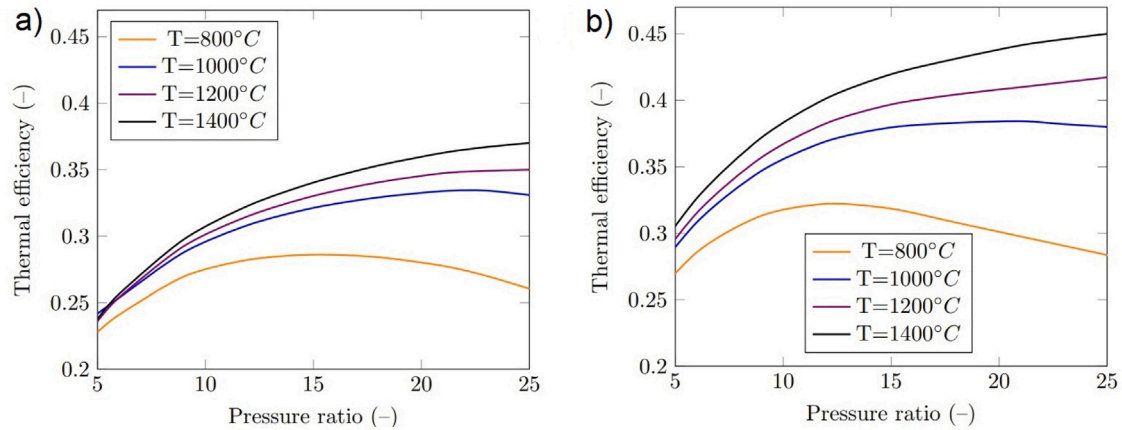


Fig. 7. Thermal efficiency of simple ammonia-fed turbine (scheme a) (a) and ammonia-fed chemically recuperated gas turbine (scheme b) (b) as a function on pressure ratio and turbine inlet temperature.

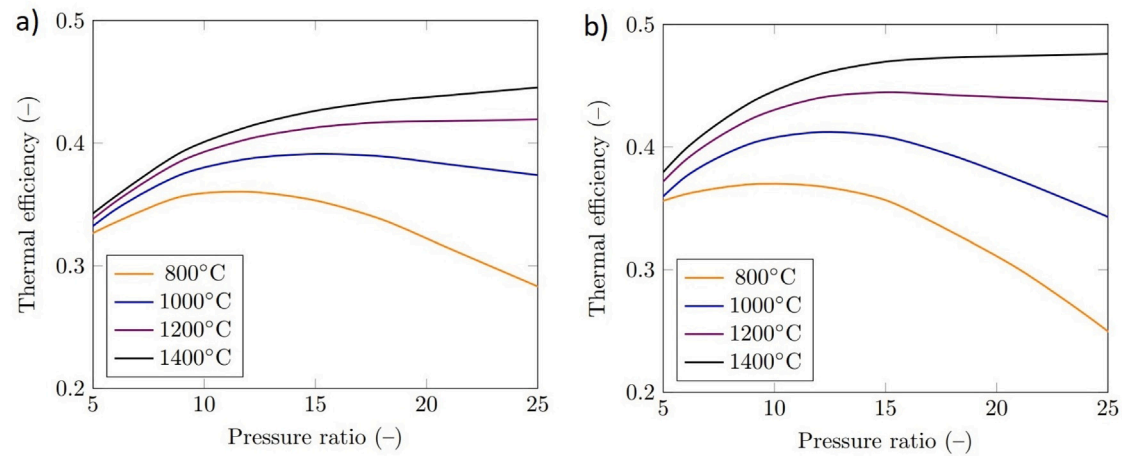


Fig. 8. Thermal efficiency for traditional combined cycle gas turbine (scheme c) (a) and chemically recuperated gas turbine with steam turbine (scheme d) (b).

3.3. Combustion performances

One of the main challenges with using ammonia in existing gas turbines is its low flame speed. This can result in incomplete combustion and reduced combustion efficiency. To address this issue, the combustion chamber volume needs to be increased to ensure complete combustion. Additionally, ammonia decomposition before combustion changes the gas turbine fuel composition. As shown in Fig. 6b, at temperatures above 500 °C, the hydrogen-rich fuel primarily consists of hydrogen, which has a significantly higher flame speed than ammonia.

Flame speed is influenced by various factors such as fuel composition, initial temperature, and pressure. Chemkin-Pro is utilized to analyze the effect of fuel composition on the flame speed. Fig. 9 shows the computational scheme from Chemkin-Pro for the laminar flame speed calculation. The flame speed was calculated for various composition of ammonia–nitrogen–hydrogen blends. The results from Chemkin-Pro are compared with data from other studies [48,49], showing good correlation. Even mixtures with low H₂ concentrations exhibit suitable flame speeds for gas turbines. Fig. 9 shows that the use of hydrogen-rich gas obtained via complete ammonia decomposition instead of pure ammonia is increasing the flame speed in combustion chamber from 5–7 cm/s to 35–40 cm/s. It should be noted that the most of existing natural gas fired gas turbine operate under flame speed about 30–40 cm/s.

Fig. 9 clearly shows the synergy of ammonia reforming to produce hydrogen and gas turbine exhaust heat recuperation. Such synergy

increases flame speed in the combustion leading to more efficient and stable combustion.

4. Future research

The thermodynamic analysis presented in this study establishes the foundational viability and performance potential of integrating exhaust-driven ammonia reforming into a gas turbine combined cycle. However, this work represents the initial step in the research pathway, and several critical areas have been identified for future investigation to advance this technology toward practical implementation. Subsequent research will focus on bridging the gap between theoretical potential and engineered system reality. First, a critical next step involves moving beyond equilibrium assumptions to develop a robust kinetic model of the ammonia decomposition process. This model will incorporate the effects of catalyst type, loading, and deactivation mechanisms to accurately predict hydrogen yield and reformer efficiency under dynamic, real-world operating conditions. This will enable the optimization of reactor design, including sizing and catalyst selection, to maximize conversion while minimizing cost and volume.

Concurrently, a comprehensive techno-economic analysis (TEA) is essential to assess the commercial feasibility of the proposed system. This research will provide detailed capital expenditure (CAPEX) estimates for key components such as the high-pressure reformer, catalyst, and specialized heat exchangers, alongside an evaluation of operational expenditures (OPEX) related to catalyst replacement and maintenance.

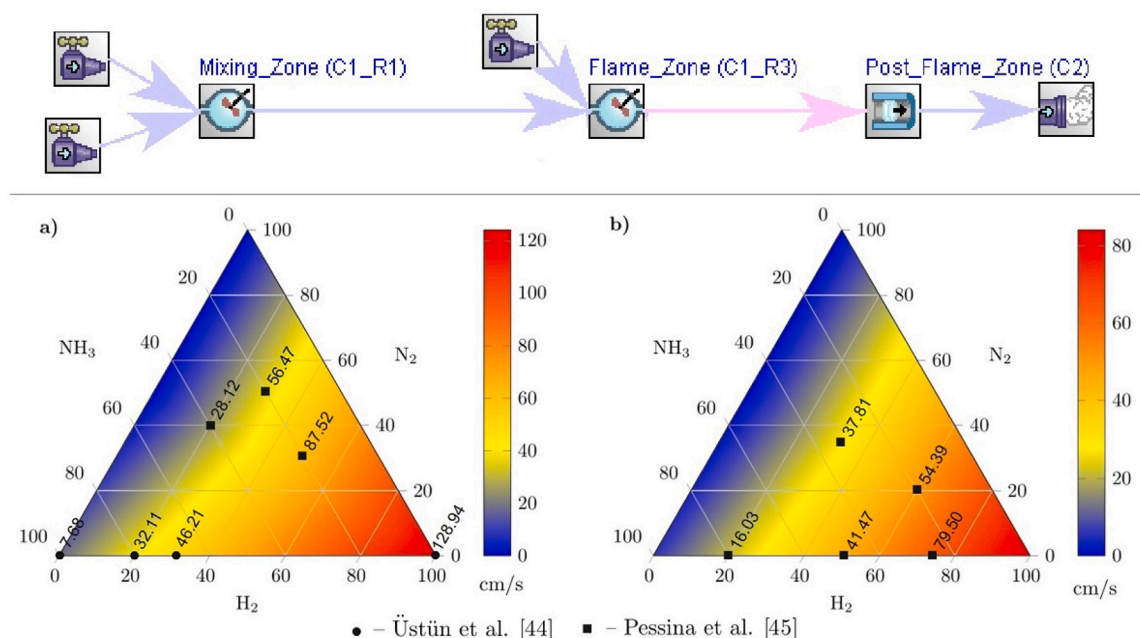


Fig. 9. Laminar flame speed for $NH_3:H_2:N_2$ blends: (a) $p = 10$ bar; (b) $p = 20$ bar.

The TEA will calculate leveled cost of electricity (LCOE) metrics to compare the economic competitiveness of the chemically recuperated cycle against other low-carbon power generation technologies.

Furthermore, building upon the thermodynamic foundation established in this study, future work will involve a detailed exergy analysis to formally quantify the reduction in exergy destruction and pinpoint the specific components where the greatest efficiency gains are realized. This will provide a deeper understanding of the “energy upgrade” achieved through thermochemical recuperation.

Additionally, to address the path toward technological implementation, research will expand to evaluate different technological alternatives for the onboard reformer, including a comparative analysis of reactor configurations (e.g., fixed-bed, microchannel) and catalyst types. This will be coupled with the previously mentioned kinetic modeling and CFD analysis to determine optimal reactor volume and catalyst mass for specific operational powers and dynamic conditions.

Moreover, the dynamic performance and control strategies of the integrated system must be thoroughly investigated. Future work will involve developing a dynamic model to simulate the system’s response to transient and load-following conditions, characterizing the thermal inertia of the reformer and its impact on fuel composition and turbine operation. This will be used to design and validate advanced control algorithms necessary for stable operation under varying load demands.

Finally, an extensive experimental program is required to validate the theoretical findings. Research will progress to the design, construction, and testing of a bench-scale reformer unit and eventually a pilot-scale gas turbine test rig. This experimental phase will be crucial for obtaining real-world data on combustion stability, NO_x and ammonia slip emissions from hydrogen–nitrogen combustion, and for verifying the overall system efficiency and operational envelope under real-world conditions.

5. Conclusion

A concept of on-board hydrogen production via ammonia decomposition reaction using exhaust heat was considered for an ammonia-fired gas turbine. Synergy of ammonia reforming to produce hydrogen and gas turbine exhaust heat recuperation is shown in part of thermal and combustion efficiency. The performance of a chemically recuperated combined cycle gas turbine (CRCCGT) was compared with that of a

conventional combined cycle gas turbine (CCGT). The CRCCGT scheme had a thermal efficiency by up to 4.7% higher than the conventional CCGT. The optimal pressure ratio for CRCCGT is 15.8 for a gas turbine inlet temperature of 1200 °C, which is 2.1 less than the optimal pressure ratio of a conventional CCPP. Changes in heat balance explain an increase in the thermal efficiency of CRCCGT: the shares of work output from gas and steam turbine cycles for CRCCGT are approximately 80%/20%, respectively, while the same shares for a conventional CCPP are approximately 75%/25%, respectively. The laminar flame speed was determined via Chemkin-Pro. It was established that under gas turbine operating conditions, laminar flame speeds ranging from 20 to 30 cm/s for a $N_2:H_2 = 1:3$ blend, representing complete ammonia decomposition into hydrogen-rich gas. It should be expected the volume of combustion chamber of existing natural gas fired gas turbines can be remained unchanged. Thus, the proposed CRCCGT scheme has two significant advantages: higher thermal efficiency and the use of hydrogen-rich gas that has better combustion efficiency comparing to ammonia. The on-board hydrogen production using gas turbine exhaust heat leads to replace of the fuel fed into combustion chamber from ammonia to hydrogen-rich gas. Therefore, in our future research the effect of fuel change on the combustion efficiency will be considered.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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